EXP 2:

Design and Performance Analysis of Chebyshev Type-I IIR Low-Pass and High-Pass Filters Using MATLAB

Objective 1:

To design Chebyshev Type-I IIR low-pass and high-pass filters for specified digital frequency requirements.

CODE:

```
clc;

clear;

close all;

Fs = 1000;

Fp = 100;

Fst = 150;

Rp = 1;

Rs = 40;

Wp1 = Fp/(Fs/2);

Ws1 = Fst/(Fs/2);

[Nlp,Wnlp] = cheb1ord(Wp1,Ws1,Rp,Rs);

[blp,alp] = cheby1(Nlp,Rp,Wnlp,'low');

Wp2 = Fst/(Fs/2);

Ws2 = Fp/(Fs/2);

[Nhp,Wnhp] = cheb1ord(Wp2,Ws2,Rp,Rs);

[bhp,ahp] = cheby1(Nhp,Rp,Wnhp,'high');

figure;

subplot(2,1,1);

freqz(blp,alp,1024,Fs);

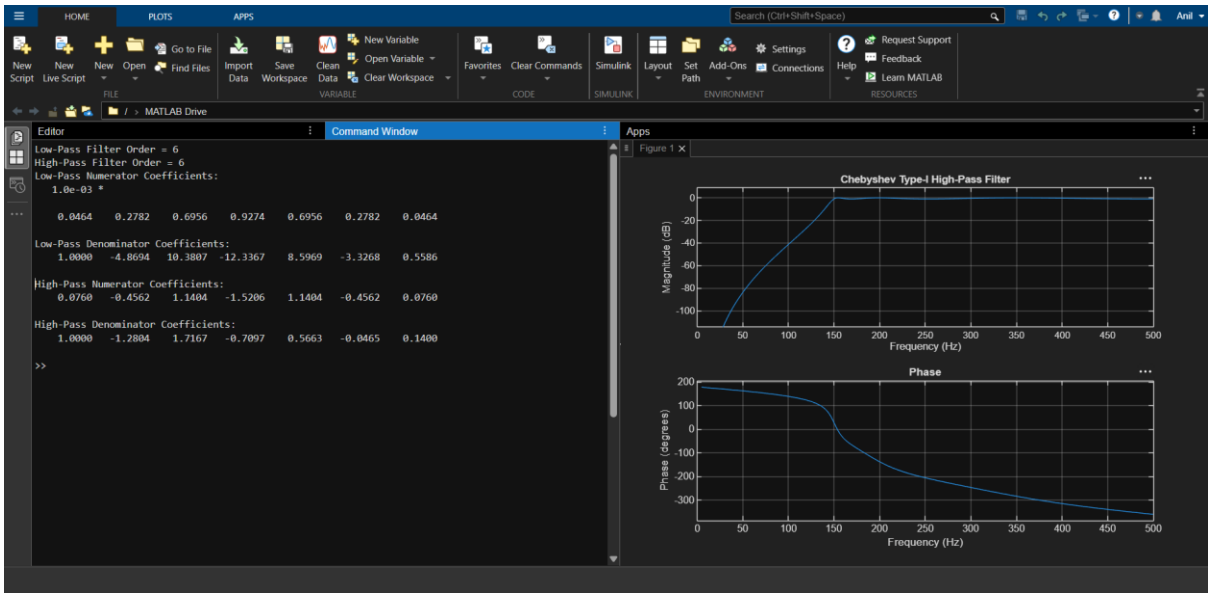
title('Chebyshev Type-I Low-Pass Filter');

subplot(2,1,2);

freqz(bhp,ahp,1024,Fs);
```

```
title('Chebyshev Type-I High-Pass Filter');
disp(['Low-Pass Filter Order = ',num2str(Nlp)]);
disp(['High-Pass Filter Order = ',num2str(Nhp)]);
disp('Low-Pass Numerator Coefficients:');
disp(blp);
disp('Low-Pass Denominator Coefficients:');
disp(alp);
disp('High-Pass Numerator Coefficients:');
disp(bhp);
disp('High-Pass Denominator Coefficients:');
disp(ahp);
```

OUTPUT:



Objective 2:

To study the effect of passband ripple on filter order and frequency response

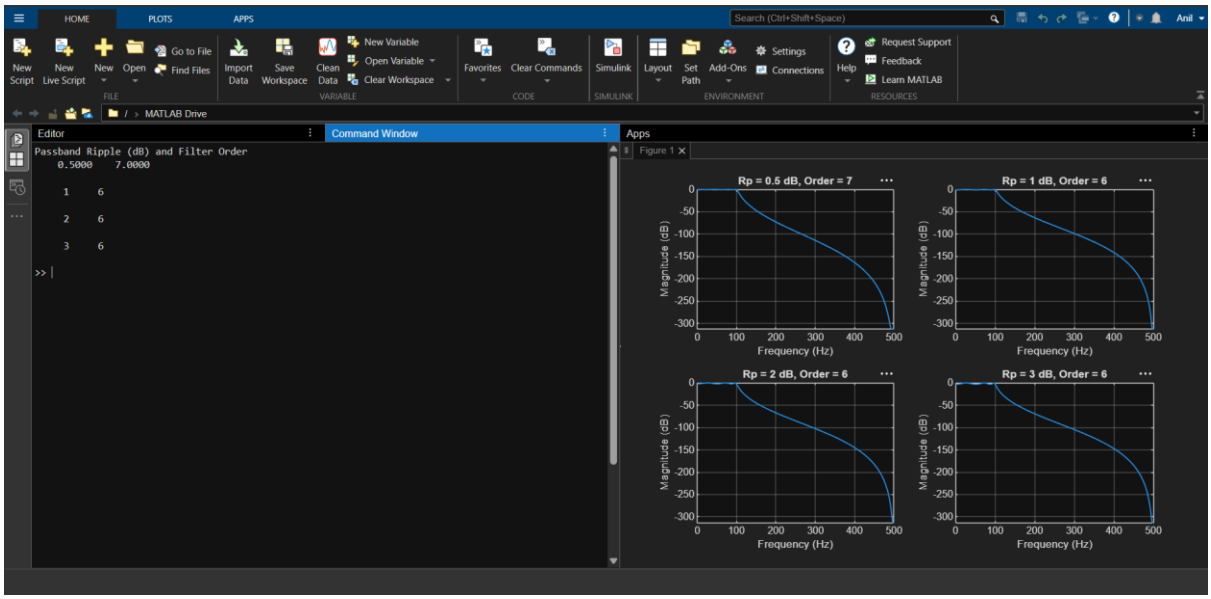
CODE:

```
clc;
clear;
close all;
```

```
Fs = 1000;
Fp = 100;
Fst = 150;
Rs = 40;
Rp = [0.5 1 2 3];
figure;
for i = 1:length(Rp)
  Wp = Fp/(Fs/2);
  Ws = Fst/(Fs/2);
  [N,Wn] = cheb1ord(Wp,Ws,Rp(i),Rs);
  [b,a] = cheby1(N,Rp(i),Wn,'low');
  [H,f] = freqz(b,a,1024,Fs);
  subplot(2,2,i);
  plot(f,20*log10(abs(H)+eps),'LineWidth',1.2);
  grid on;
  title(['Rp = ',num2str(Rp(i)),' dB, Order = ',num2str(N)]);
  xlabel('Frequency (Hz)');
  ylabel('Magnitude (dB)');
end
disp('Passband Ripple (dB) and Filter Order');

for i = 1:length(Rp)
  [N,~] = cheb1ord(Fp/(Fs/2),Fst/(Fs/2),Rp(i),Rs);
  disp([Rp(i),N]);
end
```

OUTPUT



Objective 3:

To obtain and analyze filter coefficients, magnitude and phase responses.

CODE:

```
clc;

clear;

close all;

Fs = 1000;

Fp = 100;

Fst = 150;

Rp = 1;

Rs = 40;

[Nlp,Wnlp] = cheb1ord(Fp/(Fs/2),Fst/(Fs/2),Rp,Rs);

[blp,alp] = cheby1(Nlp,Rp,Wnlp,'low');

[Nhp,Wnhp] = cheb1ord(Fst/(Fs/2),Fp/(Fs/2),Rp,Rs);

[bhp,ahp] = cheby1(Nhp,Rp,Wnhp,'high');

disp('Low-Pass Filter Coefficients');

disp('Numerator:');
```

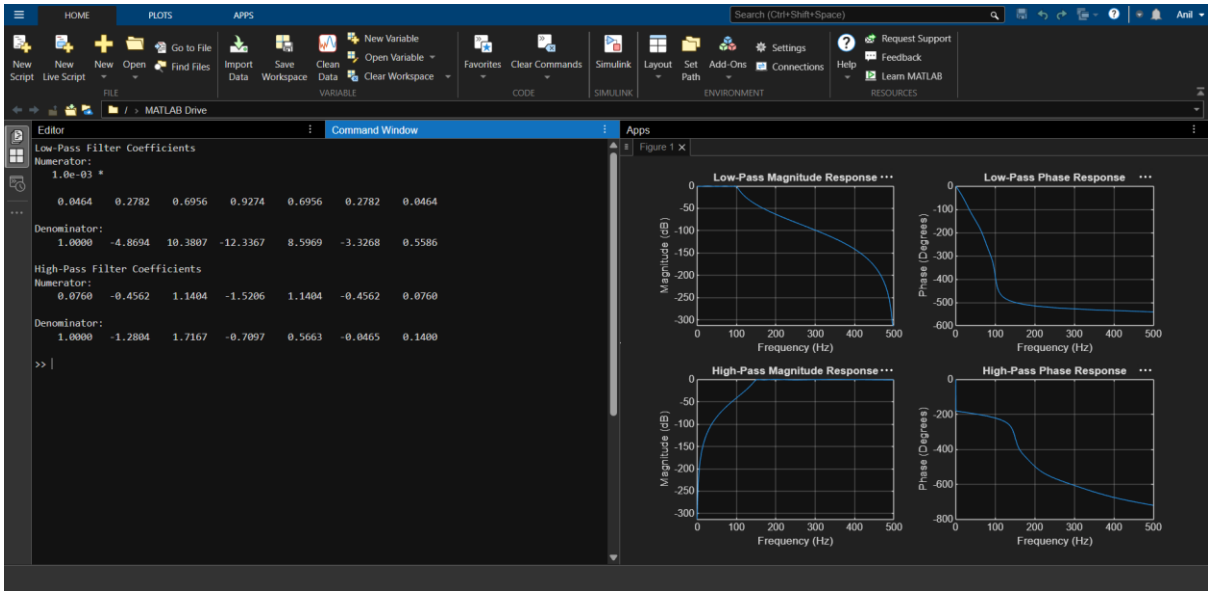
```
disp(blp);
disp('Denominator:');
disp(alp);
disp('High-Pass Filter Coefficients');
disp('Numerator:');
disp(bhp);
disp('Denominator:');
disp(ahp);
[Hlp,f] = freqz(blp,alp,1024,Fs);
[Hhp,~] = freqz(bhp,ahp,1024,Fs);
figure;
subplot(2,2,1);
plot(f,20*log10(abs(Hlp)+eps));
grid on;
title('Low-Pass Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
subplot(2,2,2);
plot(f,unwrap(angle(Hlp))*180/pi);
grid on;
title('Low-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');

subplot(2,2,3);
plot(f,20*log10(abs(Hhp)+eps));
grid on;
title('High-Pass Magnitude Response');
```

```
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

subplot(2,2,4);
plot(f,unwrap(angle(Hhp))*180/pi);
grid on;
title('High-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
```

OUTPUT



Objective 4:

To compare the Chebyshev Type-I response with the corresponding Butterworth response.

CODE:

```
clc;
clear;
```

```
close all;
Fs = 1000;
Fp = 100;
Fst = 150;
Rp = 1;
Rs = 40;
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);
[Nb,Wnb] = buttord(Wp,Ws,Rp,Rs);
[bb,ab] = butter(Nb,Wnb,'low');
[Nc,Wnc] = cheb1ord(Wp,Ws,Rp,Rs);
[bc,ac] = cheby1(Nc,Rp,Wnc,'low');
[Nbh,Wnbh] = buttord(Ws,Wp,Rp,Rs);
[bbh,abh] = butter(Nbh,Wnbh,'high');
[Nch,Wnch] = cheb1ord(Ws,Wp,Rp,Rs);
[bch,ach] = cheby1(Nch,Rp,Wnch,'high');
[Hbl,f] = freqz(bb,ab,1024,Fs);
[Hcl,~] = freqz(bc,ac,1024,Fs);
[Hbh,~] = freqz(bbh,abh,1024,Fs);
[Hch,~] = freqz(bch,ach,1024,Fs);
figure;
subplot(2,1,1);
plot(f,20*log10(abs(Hbl)+eps),'b','LineWidth',1.4);
hold on;
plot(f,20*log10(abs(Hcl)+eps),'r--','LineWidth',1.4);
grid on;
title('Low-Pass Filter Comparison');
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
legend('Butterworth','Chebyshev Type-I');
subplot(2,1,2);
plot(f,20*log10(abs(Hbh)+eps),'b','LineWidth',1.4);
hold on;
plot(f,20*log10(abs(Hch)+eps),'r--','LineWidth',1.4);
grid on;
title('High-Pass Filter Comparison');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
legend('Butterworth','Chebyshev Type-I');
disp(['Butterworth LP Order = ',num2str(Nb)]);
disp(['Chebyshev LP Order = ',num2str(Nc)]);
disp(['Butterworth HP Order = ',num2str(Nbh)]);
disp(['Chebyshev HP Order = ',num2str(Nch)]);
```

OUTPUT:

